Difference Between Data Types

Data Type	Meaning	Example
Integer (int)	Stores whole numbers without decimal point.	10, 25, -7
Float (float)	Stores numbers with decimal point.	10.5, 3.14, -2.7
String (str)	Stores text or characters inside quotes.	"Harsh", "Hello"
Boolean (bool)	Stores only two values: True or False.	True, False

Examples

Integer (int)

```
age = 18
print(age)
```

Float (float)

```
price = 99.99
print(price)
```

String (str)

```
name = "Harsh"
print(name)
```

Boolean (bool)

```
is_student = True
print(is_student)
```

2.WRITE A PYTHON PROGRAM TO CREATE THREE VARIABLES:

```
name = "Harsh"
age = 20
city = "Jaipur"
```

```python
print("Name:", name)
print("Age:", age)
print("City:", city)
```

3.Write a Python program that: Takes a user's name as input. Prints the name in uppercase. Prints the total number of characters in the name.

```python
name = "harsh"

print(name.upper())
name = "harsh"

print(name.lower())
```

Explain any five commonly used string methods in Python with examples.

1. upper()

Converts all characters of a string to uppercase.

```
name = "harsh"
print(name.upper())
```

Output:

HARSH

---

2. lower()

Converts all characters of a string to lowercase.

```
name = "HARSH"
print(name.lower())
```

Output:

harsh

---

3. replace()

Replaces one part of a string with another.

```
text = "I like Python"
print(text.replace("Python", "Java"))
```

Output:

I like Java

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4. strip()

Removes extra spaces from the beginning and end of a string.

```
text = "  Hello  "
print(text.strip())
```

Output:

Hello

---

5. split()

Splits a string into a list of words.

```
text = "Apple Mango Banana"
print(text.split())
```

Output:

['Apple', 'Mango', 'Banana']

Create a list containing the names of five fruits. • Print the complete list. • Print the first and last element. • Print the total number of items in the list.

```
fruits = ["Apple", "Mango", "Banana", "Orange", "Grapes"]


# Print complete list

print("Fruit List:", fruits)


# Print first and last element
```

```python
print("First Fruit:", fruits[0])

print("Last Fruit:", fruits[-1])


# Print total number of items

print("Total Fruits:", len(fruits))
```

6. Write a Python program to: • Create a list of numbers [10, 20, 30, 40, 50] • Add 60 to the list. • Remove 20 from the list. • Print the updated list.

```python
numbers = [10, 20, 30, 40, 50]

# Add 60
numbers.append(60)

# Remove 20
numbers.remove(20)

# Print updated list
print("Updated List:", numbers)
```

Output

Updated List: [10, 30, 40, 50, 60]

7. What is Artificial Intelligence (AI)? Explain its importance and mention any four real-life applications of AI.

Definition:
Artificial Intelligence (AI) is a branch of computer science that enables machines and computers to perform tasks that normally require human intelligence, such as learning, reasoning, problem-solving, and decision-making.

Importance of AI

1. Reduces human effort and saves time.
2. Increases accuracy and efficiency.
3. Helps in making quick decisions.
4. Automates repetitive tasks.
5. Improves productivity in various fields.

Real-Life Applications of AI

1. **Virtual Assistants** – Examples: Google Assistant and Siri.
2. **Healthcare** – Used for disease diagnosis and medical analysis.
3. **Online Shopping** – Recommends products based on user preferences.
4. **Self-Driving Cars** – AI helps vehicles detect roads, traffic, and obstacles.

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8. Identify whether the following are examples of AI and explain

why: • ChatGPT • Google Maps route prediction • Calculator • Netflix recommendations • Voice assistants (Alexa/Siri)

Example	AI or Not AI?	Reason
ChatGPT	**AI**	It understands questions, generates answers, and learns from large amounts of data.
Google Maps Route Prediction	**AI**	It analyzes traffic, distance, and user data to predict the best route.
Calculator	**Not AI**	It only performs fixed mathematical calculations and does not learn or make decisions.
Netflix Recommendations	**AI**	It studies viewing history and suggests movies or shows based on user preferences.
Voice Assistants (Alexa / Siri)	**AI**	They understand voice commands, answer questions, and perform tasks intelligently.